

7-(MathAA/10_HL_Winter_2021_Q7)-Quadratics

[Maximum mark: 7]

The equation $3px^2 + 2px + 1 - p$ has two real, distinct roots.

(a) Find the possible values for p . [5]

(b) Consider the case when $p = 4$. The roots of the equation can be expressed in the form $x = \frac{a \pm \sqrt{13}}{6}$, where $a \in \mathbb{Z}$. Find the value of a . [2]



1-(MAT/20_HL_Winter_2013_Q3)-Graphs, Exponentials-Logarithms

Consider $f(x) = \ln x - e^{\cos x}$, $0 < x \leq 10$.

- (a) Sketch the graph of $y = f(x)$, stating the coordinates of any maximum and minimum points and points of intersection with the x -axis.

[5]

- (b) Solve the inequality $\ln x \leq e^{\cos x}$, $0 < x \leq 10$.

[2]



2-(MAT/20_HL_Winter_2013_Q13)-Functions-Roots,Exponentials-Logarithms,Integration

A function f is defined by $f(x) = \frac{1}{2}(e^x + e^{-x})$, $x \in \mathbb{R}$.

- (a) (i) Explain why the inverse function f^{-1} does not exist.
- (ii) Show that the equation of the normal to the curve at the point P where $x = \ln 3$ is given by $9x + 12y - 9\ln 3 - 20 = 0$.
- (iii) Find the x -coordinates of the points Q and R on the curve such that the tangents at Q and R pass through $(0, 0)$. [14]
- (b) The domain of f is now restricted to $x \geq 0$.
- (i) Find an expression for $f^{-1}(x)$.
- (ii) Find the volume generated when the region bounded by the curve $y = f(x)$ and the lines $x = 0$ and $y = 5$ is rotated through an angle of 2π radians about the y -axis. [8]



3-(MAT/22_HL_Summer_2016_Q3)-Exponentials-Logarithms

[Maximum mark: 6]

Solve the simultaneous equations

$$\ln \frac{y}{x} = 2$$

$$\ln x^2 + \ln y^3 = 7.$$

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4-(MAT/21_HL_Summer_2016_Q7)-Exponentials-Logarithms

It has been suggested that in rowing competitions the time, T seconds taken to complete a 2000 m race can be modelled by an equation of the form $T = aN^b$, where N is the number of rowers in the boat and a and b are constants for rowers of a similar standard.

To test this model the times for the finalists in all the 2000 m men's races at a recent Olympic games were recorded and the mean calculated.

The results are shown in the following table for $N = 1$ and $N = 2$.

N	T (seconds)
1	420.65
2	390.94

- (a) Use these results to find estimates for the value of a and the value of b . Give your answers to five significant figures. [4]
- (b) Use this model to estimate the mean time for the finalists in an Olympic race for boats with 8 rowers. Give your answer correct to two decimal places. [1]

It is now given that the mean time in the final for boats with 8 rowers was 342.08 seconds.

- (c) Calculate the error in your estimate as a percentage of the actual value. [1]
- (d) Comment on the likely validity of the model as N increases beyond 8. [2]

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5-(MAT/22_HL_Summer_2017_Q6)-Exponentials-Logarithms

[Maximum mark: 5]

Given that $\log_{10} \left(\frac{1}{2\sqrt{2}} (p + 2q) \right) = \frac{1}{2} (\log_{10} p + \log_{10} q)$, $p > 0$, $q > 0$, find p in terms of q .



6-(MAT/21_HL_Summer_2019_Q11)-Exponentials-Logarithms

Consider the equation $x^5 - 3x^4 + mx^3 + nx^2 + px + q = 0$, where $m, n, p, q \in \mathbb{R}$.
The equation has three distinct real roots which can be written as $\log_2 a$, $\log_2 b$ and $\log_2 c$.
The equation also has two imaginary roots, one of which is di where $d \in \mathbb{R}$.

- (a) Show that $abc = 8$. [5]

The values a , b , and c are consecutive terms in a geometric sequence.

- (b) Show that one of the real roots is equal to 1. [3]

- (c) Given that $q = 8d^2$, find the other two real roots. [9]



7-(MAT/10_HL_Winter_2020_Q5)-Exponentials-Logarithms, Sequences-Series

[Maximum mark: 5]

The first term in an arithmetic sequence is 4 and the fifth term is $\log_2 625$.

Find the common difference of the sequence, expressing your answer in the form $\log_2 p$, where $p \in \mathbb{Q}$.



8-(MathAA/10_HL_Winter_2021_Q3)-Exponentials-Logarithms

[Maximum mark: 5]

Solve the equation $\log_3 \sqrt{x} = \frac{1}{2\log_2 3} + \log_3(4x^3)$, where $x > 0$.



1-(MAT/21_HL_Summer_2016_Q5)-Graphs

The function f is given by $f(x) = \frac{3x^2+10}{x^2-4}$, $x \in \mathbb{R}$, $x \neq 2$, $x \neq -2$.

- (a) Prove that f is an even function. [2]
- (b) (i) Sketch the graph $y = f(x)$.
- (ii) Write down the range of f . [5]

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